NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

SSCS, Midterm Examination Spring 2026

Time: 90 minutes | **Course Title (MT-272) – B** | **Max Marks: 20**

Note: Attempt all questions and show relevant working. Write your solutions neatly and organize them by question number.

[Marks]

Q1) [CLO-1] a) Find the curved area of the frustum formed by rotating the segment of the line $y = 2x + 1$, between $1 \leq x \leq 3$, about the $x - axis$. [04]

b) The points A, B, and C have position vectors $\vec{a} = 2i + j + k$, $\vec{b} = i + 3j - k$, and $\vec{c} = 3i - j + 2k$, respectively, relative to the origin **O**. Find the vector equation and equation of the plane passing through these points. [03]

c) A student wants to use eggs and milk to increase the amount of protein and calcium in his daily diet. An ounce of eggs contains 4 grams of protein and 10 milligrams of calcium, while an ounce of milk contains 2 grams of protein and 30 milligrams of calcium. How many ounces of eggs and milk should the student consume each day to obtain exactly 56 grams of protein and 620 milligrams of calcium? Form the system of linear equations and write it in matrix $\boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b}$. [03]

Q2) [CLO-2] a) Determine the values of λ and μ for which the system has no solutions, exactly one solution, or infinitely many solutions. [05]

$x + y + z = 3$ $\qquad 2x + 3y + z = 7$ $\qquad 3x + 4y + \lambda z = \mu$

b) In continuation of above question, determine the general solution in case of infinitely many solutions. [02]